CECH COHOMOLOGY OF PRESHEAVES COMPUTES THE RIGHT DERIVED FUNCTORS OF H^0-CHECK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-etale-cohomology/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The same-chapter references are located at their \label line inside the pinned chapter file; the cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- Cech complex and Cech cohomology of a family in a presheaf of abelian groups. Same chapter, Definition environment labelled definition-cech-complex (source line 1477, Stacks tag 03OL); the statement opens by referring to its notation and assumptions. Supplies the functors H^p-check(U, -) on PAb(C); the comparison with derived functors is the counted content.
- Universal delta-functors and effaceability. Standard homological algebra; the proof uses that an effaceable delta-functor is universal and that two universal delta-functors agreeing in degree 0 agree everywhere. Supplies the two general facts combined in the proof; the identification of effaceability with the previous lemma is the counted step.

Source reference. Original definition 03OL: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 1477–1477. Same-chapter pointer named in the first line of the counted statement (line 1725) as the source of notation and assumptions; the comparison of delta-functors is retained verbatim in the counted span.

Source reference. Original lemma 03OR: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 1702–1702. Same-chapter pointer invoked at proof line 1731 for effaceability of the Cech functors; it supplies that input only, while the universality and degree-zero comparison are retained in the counted span.

Source reference. Original lemma 03AU: sources/stacks-chow/tags.txt, source lines 4323–4323. Cross-chapter 'for more details' pointer at proof lines 1736-1738, invoked after the argument is complete; the retained proof is self-contained in the two delta-functor facts.

Theorem 0.1. *Notation and assumptions as in Definition 03OL. On PAb($\mathcal{C}$) the functors $\check{H}^p(\mathcal{U}, -)$ are the right derived functors of $\check{H}^0(\mathcal{U}, -)$.*

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Proof. By the Lemma 03OR, the functors $\check{H}^p(\mathcal{U}, -)$ are universal δ-functors since they are effaceable. So are the right derived functors of $\check{H}^0(\mathcal{U}, -)$. Since they agree in degree 0, they agree by the universal property of universal δ-functors. For more details see Cohomology on Sites, Lemma 03AU. □